

PRAVEENA SATTI

AI/ML RESEARCHER | TRUSTWORTHY AI | INFORMATION RETRIEVAL | MEDICAL & MULTIMODAL ML

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RESEARCH PROFILE

AI/ML researcher and engineer focused on trustworthy and reproducible learning systems across information retrieval, language models, medical image segmentation, multimodal/time-series learning, and production ML. Hands-on work spans Transformer NLP, RAG and hybrid retrieval, state-space modeling, graph systems, cloud ML pipelines, checkpoint verification, distribution-shift stress testing, and verification-aware agent workflows. Published work includes BERT-based multi-label emotion classification and an open technical preprint on backward-consistent multimodal reconciliation.

Research objective: Develop AI systems whose predictions and actions can be evaluated, traced to evidence, reproduced reliably, and adapted safely under real-world distribution shift.

RESEARCH INTERESTS & TECHNICAL DEPTH

Research: Trustworthy/Reliable AI | Medical Image Segmentation | Test-Time Adaptation | Distribution Shift | Information Retrieval & RAG | NLP & Transformers | Multimodal & Time-Series Learning | Agentic AI | Graph ML | ML Systems

Methods: PVTv2/EMCAD | Dice/Jaccard/HD95/ASD | uncertainty & entropy | reliability gating | BM25 | sparse/dense retrieval | SPLADE | Lucene | BERT/Transformers | state-space models & RTS smoothing | semantic search | graph reasoning

RESEARCH REPRODUCIBILITY & OPEN-SOURCE CONTRIBUTIONS

EMCAD Synapse Checkpoint Verification & ReliEMCAD Research Extension 2026

- Independently verified the authors' released PVT-EMCAD-B2 checkpoint on the official 12-volume Synapse test split using CPU-compatible batched inference; observed 83.86% mean Dice and 74.68% mean Jaccard, closely matching paper references of 83.63% and 74.65%.
- Recorded the exact official-repository commit, checkpoint SHA-256, model parameter count, environment details, raw per-case/per-class metrics, and execution logs; checkpoint loaded with 0 missing and 0 unexpected keys. Reported HD95 20.57 versus paper reference 15.68 without masking the reproducibility gap.
- Proposed ReliEMCAD: reliability-gated lightweight test-time adaptation for medical segmentation under domain shift. In a 148-slice Synapse volume pilot, a controlled gamma intensity shift reduced Dice from 72.59% to 53.78% and Jaccard from 63.64% to 46.62%, motivating uncertainty-aware freeze/adapt/referral and rollback decisions.

Castorini - Foundations of Retrieval | Anserini & Pyserini 2026

- Reproduced retrieval experiments across classical, sparse, and dense settings and contributed verified reproduction-log results to Castorini's official information-retrieval toolkits.
- Both contributions passed maintainer review and were merged upstream: Anserini PR #3391 and Pyserini PR #2650; work covered BM25, Lucene retrieval, SPLADE-family retrieval, neural/dense retrieval, benchmark reproduction, and metric verification.

RESEARCH & PUBLICATIONS

Emotion Classification Using BERT: A Comprehensive Study 2023

Journal of Propulsion Technology, 44(4) - peer-reviewed publication

- Developed a Transformer-based multi-label emotion-classification workflow on GoEmotions covering preprocessing, BERT fine-tuning, evaluation, and label-level analysis.

BioTwin-PE: Multimodal Placental Monitoring with Backward-Consistent Smoothing 2026

Open-access technical preprint

- Proposed APRE and TPSM for reconciling asynchronous evidence with continuous wearable signals using backward-consistent temporal reasoning and state-memory concepts.

RESEARCH-RELEVANT EXPERIENCE

AI / ML Engineer Intern - Infor, Hyderabad 2025-Present

- Built end-to-end retrieval-augmented generation pipelines using LangChain, vector indexing, SQL preprocessing, and REST deployment layers for enterprise knowledge retrieval.
- Developed NLP automation modules reported to reduce manual workflow processing time by 30%, and designed reusable AWS data/ML workflows using S3, Lambda, Docker, CI/CD, monitoring, and preprocessing components.

Software Development Intern - Swecha Internship

- Improved backend reliability for an open-access learning platform serving 5,000+ localized learners by auditing API/database retrieval paths, resolving bottlenecks, and building reusable backend modules; reported a 15% execution-efficiency improvement.

SELECTED RESEARCH & ENGINEERING PROJECTS

ReliEMCAD - Reliability-Gated Test-Time Adaptation for Medical Segmentation

- Designed a lightweight deployment-time framework that estimates distribution shift and predictive uncertainty, then chooses freeze, selective adaptation, or referral rather than adapting every target input unconditionally.
- Planned selective BN/affine or low-rank updates with entropy + consistency + anchor objectives, acceptance testing, rollback, calibration, and harmful-adaptation tracking.
- Evaluation plan covers Dice/Jaccard/HD95/ASD, ECE, uncertainty-error AUROC, selective Dice/coverage, latency, memory, and adaptation-failure analysis.

PraxisMesh - Verified Agent Operations Platform

- Designed a safety-oriented agent execution architecture with typed task graphs, deterministic policy gates, human approval, independent postcondition verification, and tamper-evident audit trails.
- Focused the system on explicit execution boundaries and evidence-producing verification rather than unconstrained autonomous action.

Clinical Trial Matching Engine - Retrieval / NLP

- Implemented hybrid semantic + structured retrieval over public ClinicalTrials.gov metadata using sentence-transformers, Chroma, FastAPI, and Docker.
- Structured the pipeline for evidence-grounded candidate-to-trial retrieval with reusable preprocessing and API layers.

Multimodal Vital-Signs Anomaly Detector - State-Space ML

- Built quality-aware state-space fusion with RTS smoothing for missing/noisy HR, SpO2, respiration, and temperature streams.
- Modeled uncertainty and irregular longitudinal observations rather than assuming complete, uniformly sampled physiological data.

Repo-Aware Coding Assistant - Code Intelligence

- Built Python AST chunking, call-graph expansion, and lexical retrieval for grounded file/line search and repository-aware change suggestions.
- Designed outputs around traceable code locations and dependency context to reduce unsupported repository edits.

Additional Research/Engineering Systems

Natural-Language-to-Infrastructure Generator | Hybrid Wiki Search Engine | Transaction Graph Fraud Detector | Distributed ML Task Queue | ETL Observability Dashboard | HireLens ATS analysis | Avenkin care-coordination platform.

EDUCATION

Bachelor of Technology (Honours) - Computer Science & Engineering (AI/ML)

2022-2026

KL University

- Specialization: Artificial Intelligence and Machine Learning. Academic work and independent projects span machine learning, deep learning, NLP, information retrieval, computer vision/medical imaging, cloud/data systems, and software engineering.

RESEARCH WORKFLOW & EVALUATION PRACTICE

- Problem formulation -> related-work review -> baseline selection -> reproducible experiment setup -> metric validation -> error/failure analysis -> documented evidence.
- For ML reproducibility, record dataset/split, source revision, checkpoint provenance, environment, raw metrics, and limitations; distinguish checkpoint verification from independent retraining or multi-run replication.
- For trustworthy AI, prefer explicit uncertainty, calibration, acceptance tests, rollback, grounded evidence, controlled execution, and documented failure modes over unsupported model claims.

TECHNICAL SKILLS

ML/AI: Python | PyTorch | TensorFlow | Scikit-learn | Transformers/BERT | sentence-transformers | LangChain | information retrieval | semantic search | state-space modeling | graph ML

Data/Systems: Pandas | NumPy | PySpark | SQL | FastAPI | Flask | Docker | Linux | Neo4j | NetworkX | REST APIs | CI/CD

Cloud/MLOps: AWS S3/Lambda | Terraform | monitoring | reproducible environments | checkpoint/data provenance | experiment logs

SELECTED CREDENTIALS & PROFESSIONAL AFFILIATIONS

Certifications: AWS Certified Cloud Practitioner | Google Associate Cloud Engineer | GitHub Foundations | Neo4j Certified Professional | Automation Anywhere Advanced RPA Professional

Affiliations: IEEE Member | IEEE Women in Engineering | IEEE SIGHT

RESEARCH EVIDENCE & CLICKABLE LINKS

- EMCAD / ReliEMCAD reproducibility code
- Castorini Anserini merged PR #3391
- Castorini Pyserini merged PR #2650
- GitHub research portfolio
- Research portfolio website
- ORCID 0009-0000-6555-5072